

UNIVERSE Homework project

Build Your Own Planetary System - teamwork

School: **Grade: 5** **Subject: Geography** **Teacher:** **Deadline:**

Goals:

- Understand planetary systems, orbits, and gravitational interactions, connect scientific concepts with creative problem-solving.
- Develop teamwork, practice respectful listening and idea sharing, reflect on individual roles and group collaboration.
- Innovate by creating fictional planets and phenomena.

Learning outcomes:

- Design and build a 3D model, poster, or digital simulation.
- Present and explain your planetary system to peers

Instructions:

You and your team will design a fictional planetary system. Each of you will research one celestial object (planet, moon, comet, asteroid, etc.) and then combine your ideas to create a full system. Your system should include interesting scientific details, a clear structure, and a story for how your planets interact.

Steps to follow:

- Assign roles fairly: (researcher,) model builder, presenter, illustrator – designer
- Research Your Object: each of you selects one celestial object to focus on. Answer these questions:
 - What is its size, temperature, and atmosphere?
 - What is its orbit and how does it move relative to other objects?
 - Does it have special features like rings, volcanoes, or magnetic storms?
- **Combine Ideas:**
 - assemble your planetary system.
 - Decide together on the position of each planet, moons, or other objects.

- Think about interactions: gravitational effects, day/night cycles, collisions.
- **Create a Model or Visual**
 - Build a 3D model, digital simulation, or illustrated poster of your system. Make sure your model shows each planet's position and size, orbits of planets and moons and Any special phenomena
- **Get prepared** to explain the class how your system work and what makes it unique?

Drawings / Note

Reflection Notes:

- What did you learn about the Universe?
- How did you contribute to the final product? Talk about yourself and others.
- What was your favorite part of the project?
- What did you learn about teamwork and collaboration? What worked well, what was challenging?